# The cost of precision-fermented whey and egg proteins

A Humbird-style techno-economic analysis of β-lactoglobulin and ovalbumin, from stoichiometry to the physical limit

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**Status.** Draft for the author's review; not yet peer-reviewed. The model code, the evidence extraction and this draft were produced with an AI assistant (Claude, Anthropic) under the author's direction. Every number in this document is computed by the model and read from `results/analyses.json` when the document is compiled. Every input traces to a source and an evidence grade in `data/` (Appendix A). Inputs graded E are placeholders; they are listed in Section 6.

## Summary

Precision fermentation (PF) uses an engineered microbe to secrete a single animal protein. It is the main route to animal-free dairy and egg proteins. We estimate what food-grade β-lactoglobulin (BLG, the main whey protein) and ovalbumin (OVA, the main egg-white protein) cost to make, and what performance would bring them to price parity. The method is Humbird's (2021): element- and electron-balanced black-box stoichiometry, an oxygen-transfer-limited fed-batch fermenter, and a capital estimate anchored to the market price of baker's yeast. We extend it with a secreted product and a downstream train, and calibrate it against three published designs.

The base case is a secreting **Komagataella phaffii** strain at 30 g/L in a 72 h production phase, in 200 m³ fermenters, making 10 kt/yr of protein in the US Midwest (2025 dollars). Its minimum selling price, with a capital charge of 14.6% of investment per year, is **$29.2/kg BLG** and **$31.6/kg OVA**. With the lighter capital convention used by the published PF studies it is $16.8 and $18.4. At the best public academic titers (5.1 g/L BLG, 2 g/L OVA) the price is $188 and $467/kg. At the private-sector average titer reported to GFI (42 g/L) it is $22.8–$25.2/kg. For comparison, whey protein traded at $7–$18 per kg protein in 2015–2025 (export unit values), and near $34 at the 2026 spot peak. Dried egg white traded at $8–$20.

Three results stand out:

1. **Oxygen transfer and heat removal set a titer ceiling.** A 200 m³ fermenter at 80 g/L cells reaches at most 54 g/L of BLG in a 72 h production phase. Within that wall, $20/kg needs about 38–46 g/L in 48–72 h. $10/kg is out of reach with the base capital convention.
2. **The physical-limit floor** (yield near the energetic ceiling, lowest maintenance, $O_2$-limited rate, 100 kt/yr) is $6.8 per kg BLG and $11.4 per kg OVA, or $3.5 and $6.3 with light capital.
3. **Capital conventions are the largest structural uncertainty.** Our mass and energy balances match the published designs within a few percent. Their costs are 2.7–2.8× lower than ours, because of lighter capital and fixed-cost conventions, not better biology.

EU sugar and energy prices add $4.4/kg to the base case.

| Scenario | Titer (g/L) | BLG, Humbird | BLG, light | OVA, Humbird | OVA, light |
|---|---|---|---|---|---|
| Best public academic strain | 5.1 / 2 | $188 | $103 | $467 | $230 |
| Base case (72 h production) | 30 | $29.2 | $16.8 | $31.6 | $18.4 |
| Industry benchmark (GFI private average) | 42 | $22.8 | $13.3 | $25.2 | $14.9 |
| Physical limit (100 kt/yr) | 130 / 50 | $6.8 | $3.5 | $11.4 | $6.3 |
| Incumbent, per kg protein (2015–2025) | | | whey $7–$18 | | egg white $8–$20 |

Table 1: Minimum selling price ($/kg target protein, 2025) with a capital charge of 14.6% of total capital investment per year. "Humbird" and "light" are the two capital conventions of Section 2.5. 10 kt/yr and US Midwest unless noted. Incumbent prices: export unit values per kg protein (Section 4.2).

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# 1 Question and approach

PF companies now sell BLG (Perfect Day, Remilk) and OVA (EVERY), and others are entering the market (Vivici, Onego Bio). Whether these proteins can displace animal proteins at scale, or stay a premium niche, depends on what they cost to make. Published estimates disagree by a factor of several. Company claims are rarely checked against the physics of aerobic fermentation. This study asks:

> What does it cost to make food-grade BLG and OVA by precision fermentation with today's public strain performance, and at the physical limits of aerobic fermentation? What combination of titer, rate, yield, downstream design and scale would bring them to price parity with whey and egg-white proteins?

| # | Sub-question | Answered in |
|---|---|---|
| Q1 | What do BLG and OVA cost at public (academic) and claimed (industry) performance? | Section 4.2, Figure 2 |
| Q2 | What drives the cost, and how good is the evidence for each driver? | Section 4.3, Figure 3, Figure 4, Figure 5 |
| Q3 | Which combinations of performance reach $10 and $20 per kg protein? | Section 4.4, Figure 6 |
| Q4 | What is the cost at the physical limits? | Section 4.5, Figure 7 |
| Q5 | Are published company claims consistent with $O_2$-transfer and cooling limits? | Section 4.6, Table 7 |
| Q6 | How much do siting (US vs EU) and scale (1–100 kt/yr) matter? | Section 4.7, Figure 8 |

Table 2: The six sub-questions fixed before any model output existed (`docs/SCOPE.md`).

**Why Humbird's method.** Humbird's (2021) cultured-meat analysis builds costs from conservation laws: element and electron balances, oxygen transfer, heat removal. Its capital estimate is tied to a real product price, baker's yeast at about $1.90/kg. We port his baker's-yeast workbook exactly (Section 3) and extend it with three additions: a secreted protein, a downstream train, and the specifics of **K. phaffii** and **Trichoderma reesei**.

**Rules we followed.** The analysis was run under a written honesty protocol (`PLAN.md`):

- Four scenarios are never merged: academic, industry, required and physical limit.
- Every input carries an evidence grade, A to E.
- The model structure was frozen before the calibration anchors were run. Calibration misses are recorded, not tuned away.
- Priors about the results were written down before the model existed; they are scored in Section 7.
- Model-to-model agreement is treated as a consistency check, not as validation.

# 2 Methods

## 2.1 Plant, products and functional unit

The functional unit is 1 kg of target protein in a spray-dried powder that meets the product specification. The base specification is a **concentrate**, following Remilk's GRAS notice for BLG (FDA GRN 1056): at least 70 wt% protein, at least 80% of it BLG, at most 7% moisture. OVA uses the same specification by analogy. The **isolate** variant (97–99% target protein, dry basis) adds an ammonium-sulfate precipitation, following the Intelligen (2025) OVA design.

The plant is greenfield, single-product and nth-of-a-kind. It is built to food, not pharmaceutical, good manufacturing practice. The base is 10 kt/yr of protein in the US Midwest, swept from 1 to 100 kt/yr, with a north-west Europe variant. Money is 2025 US dollars. Excluded costs: R&D, strain licences, regulatory approval, sales and distribution, land, first-of-a-kind premiums and carbon pricing. Figure 1 shows the base-case plant.

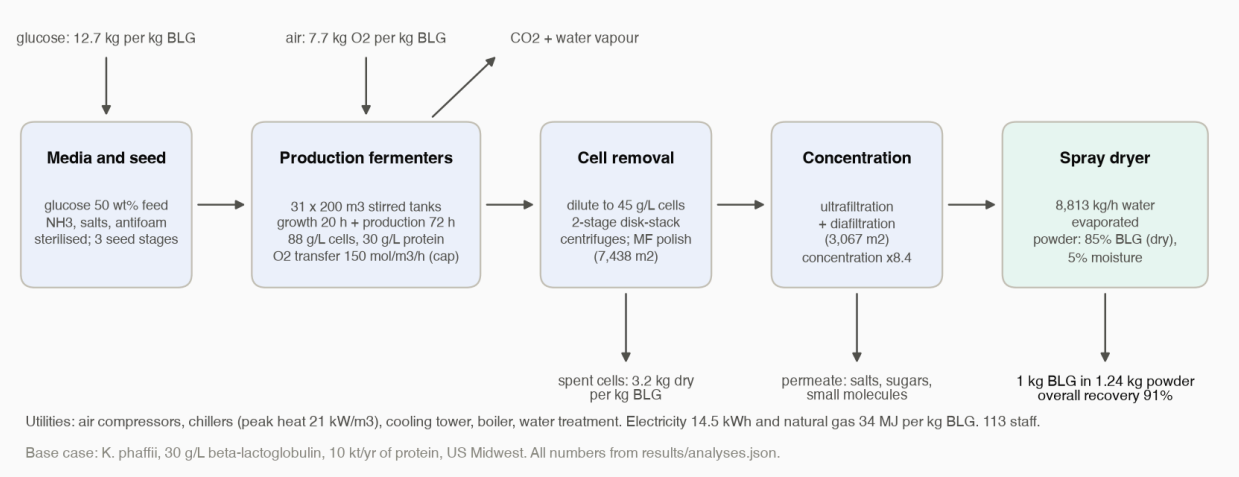



Figure 1: Base-case process: **K. phaffii** secreting BLG at 30 g/L; flows per kg of BLG.

## 2.2 Stoichiometry and energetics

Each metabolic process is written as one element-balanced "black-box" reaction on C, H, O and N: glucose + $NH_3$ + $O_2$ ⟶ biomass (or protein) + $CO_2$ + $H_2O$. Electrons are tracked through the degree of reduction γ (glucose 4, biomass 4.2, protein 4.3 per C-mol). This closes the $O_2$ and $CO_2$ balances once the carbon split is fixed.

**Growth.** Biomass yield comes from the Gibbs-energy dissipation correlation of Heijnen and van Dijken (1992), as in Humbird's workbook. The energy dissipated per C-mol of biomass formed is

$$D = 200 + 18(6 - c)^{1.8} + \exp\left\{\left[(3.8 - \gamma_s)^2\right]^{0.16}(3.6 + 0.4c)\right\} \quad \text{kJ/C-mol},$$

where $c$ and $\gamma_s$ are the carbon number and degree of reduction of the substrate. The catabolic reaction (glucose to $CO_2$) is run just enough to supply $D$. For glucose this gives a maximum biomass yield of about 0.52 g/g. The measured **K. phaffii** yield is 0.50–0.58 g/g (`data/evidence/hosts.yaml`, grade A).

**Maintenance.** Maintenance uses Tijhuis et al. (1993), $m_G = 4.5\exp\left[-69\frac{000}{R}\left(\frac{1}{T} - \frac{1}{298}\right)\right]$ kJ per C-mol biomass per hour. At 30 °C this is about 0.018 g glucose per g cells per hour. The measured **K. phaffii** range is 0.002–0.029 g/g/h (grade A). The base keeps Humbird's value.

**Product.** The protein's elemental formula comes from its mature UniProt sequence:

- BLG: $C_{817}H_{1312}N_{206}O_{248}S_9$, 18.3 kDa;
- OVA: $C_{1901}H_{2997}N_{499}O_{575}S_{22}$, 42.8 kDa.

The product reaction makes protein from glucose and ammonia. It respires enough extra glucose to pay for synthesis, so the product yield on glucose, $Y_{ps}$, fixes its $O_2$ demand. An energetic ceiling follows from counting ATP: 4 ATP per peptide bond plus amino-acid synthesis, paid by respiration at up to 28 ATP per glucose. For BLG the ceiling is about 0.6 g/g. The base yield, 0.28 g/g (Intelligen's assumption, grade B), is 46% of it. No measured BLG or OVA yield on sugar was found (Section 7, prior 3).

**Heat.** Heat release is computed from formation enthalpies. It is cross-checked against the empirical 460 kJ per mol $O_2$ (Roels; grade A in `engineering.yaml`).

> **Check (Pablo):** The dissipation formula and its units; the conversion of $m_G$ to 0.018 g/g/h; the ATP ceiling (`src/pftea/product.py`, `energetic_ceiling`). These are the derivations the whole cost structure rests on, and none has been checked by a second person.

## 2.3 The fed-batch fermenter and the oxygen wall

Each production batch has two phases:

1. **Growth** at the maximum specific rate, slowed so that the oxygen uptake rate (OUR) never exceeds what the vessel can transfer and the heat never exceeds what it can remove, up to the production cell density $X_{\text{prod}}$ (base 80 g/L; Intelligen 83 g/L).

2. **Production** for $t_{\text{prod}}$ hours, with a small residual growth and a constant specific productivity $q_p$.

The initial liquid volume is solved so that the broth ends at 80% of the vessel. $q_p$ is solved so that the harvest titer equals the target. A liquid mass balance tracks feed, $O_2$ uptake, $CO_2$ release and evaporation.

**Oxygen transfer.** The volumetric transfer coefficient follows van 't Riet's correlation for non-coalescing broths, $k_L a = 0.002(P/V)^{0.7} U_s^{0.2} 1.025^{T-20}$ ($s^{-1}$, SI units), as in Humbird's microbial fermenters. The driving force is a log-mean between the bottom (hydrostatic pressure, fresh air) and the top (back-pressure, depleted gas). For each demanded OUR, the model picks the air rate and agitation power that minimise compressor + $(1 +$ refrigeration factor$) \times$ agitator power. Agitator power ends up as heat that the chillers must remove. The transferable maximum is then capped at the design value Humbird uses, 150 mol $O_2/m^3/h$. Humbird notes that 200 mol/m³/h "is about as high as it gets" in fermentation. The heat load is capped at 25 kW/m³ (grade D; Section 6).

**The oxygen wall.** The production phase's $O_2$ demand is set by the titer and the time. If the peak demand exceeds the cap, the design is flagged **infeasible**. We never cost it as if it worked. A rough scale shows why this binds. Each kg of BLG costs about 241 mol $O_2$ across the whole batch, including cell growth and maintenance. So 30 kg/m³ of product needs about 7.2 kmol $O_2$ per m³. Spread over the 92 h batch, that is a mean of 79 mol/m³/h, against a cap of 150. The peak comes at the end of growth and is higher. Doubling the titer in the same time pushes it past the cap: at 72 h the ceiling is 54 g/L. Halving the time brings it close: at 36 h the ceiling is 34 g/L.

### 2.4 Downstream processing

The downstream train (Figure 1) follows Intelligen's OVA design, with evidence-graded performance (`data/evidence/dsp.yaml`):

- **Cell removal.** Dilute to 45 g/L cells, then two-stage disk-stack centrifuges (97% recovery).
- **Microfiltration polish** at 10 L/m²/h (EVERY patent example, grade B).
- **Ultrafiltration and diafiltration** at 30 L/m²/h. Rejection is 0.995. BLG is concentrated to 150 g/L, OVA to its practical solubility of 50 g/L. Three diafiltration volumes are used.
- **Spray dryer** at 4.87 MJ per kg of water evaporated (industry survey, grade A).
- **Isolate only:** ammonium-sulfate precipitation (350 g/L; 89% recovery), then washed basket centrifuges.

The target protein share of the concentrate powder is 85% on a dry basis (grade D; the GRN 1056 minimum is 56%). The base overall recovery is 91%.

### 2.5 Capital: two conventions

**Humbird convention (base).**

- Purchased equipment costs come from Humbird's cost curves (Aspen quotes, 2018 dollars).
- Costs are escalated to 2025 with a factor of 1.33 (CEPCI and the BLS machinery price index; grade B, range 1.23–1.52).
- Installation, buildings, engineering, fees and contingency use Humbird's "commodity aerobic" facility class: a food-grade plant with the process inside a sanitary building. The effective Lang factor (investment over equipment cost) is 5.1.
- Maintenance is 4% and insurance plus local tax 5% of total capital investment (TCI) per year.
- This convention is the one tied to a market price (anchor 1).

**Light convention (sensitivity).** This is the convention of NREL's 2011 design report, which the published PF studies resemble:

- vendor-quote fermenters, over 3× cheaper than Humbird's curve at 300 m³;
- NREL's installation factors plus indirect costs of 60% of direct cost;
- maintenance of 3% of inside-battery-limits (process-area) direct cost, and insurance of 0.7% of fixed capital.

We report both conventions throughout. Their ratio is the largest structural uncertainty in the study.

### 2.6 Operating costs, labour and money

**Operating costs.** Raw materials, utilities and consumables come from the mass and energy balance and the prices in Appendix A. Prices are grade A–B: EIA, Eurostat and BLS statistics, trade unit values and published designs.

**Labour.** Labour follows Humbird's operator-attention model. Operators are assigned per fermenter by the fraction of time a batch needs attention. Continuous areas (media, centrifuges, membranes, dryer, utilities, maintenance, QC) have fixed staffing. There are 4.5 shifts per position, a supervisor for every ten staff at 1.4× salary, and 100% overhead. The base plant employs 113 people.

**Headline metric.** The minimum selling price is $\mathrm{MSP}_{\mathrm{CCF}}$: annual operating cost plus a capital charge, divided by annual output. The charge is the capital recovery factor $i/(1-(1+i)^{-n}) = 0.146$ per year at $i = 7.5\%$ and $n = 10$ years (Humbird). It is a cost with a return on capital, not a market price. Margins, functionality and cost-in-use are outside the model.

**Other metrics.** We also compute:

- the cost of production (COP): operating cost plus depreciation, no return;
- an NREL-style discounted-cash-flow price, $\mathrm{MSP}_{\mathrm{DCF}}$: 10% IRR, 30 years, 21% tax, 7-year MACRS.

For the base BLG case these are \$25.3 (COP) and \$28.3 ($\mathrm{MSP}_{\mathrm{DCF}}$).

### 2.7 Evidence grades, scenarios and uncertainty

Every parameter and evidence item is graded (Table 3). The four scenarios combine grades differently and are never averaged:

- **academic:** best public strain performance. BLG is 5.14 g/L in **K. phaffii** (Li et al. 2026) and OVA is 2 g/L in **T. reesei** (Aro et al. 2023). Both come from abstracts; the full texts are not yet read.
- **industry:** the private-sector average titer that companies reported to GFI (2025): 42 g/L (grade C), with 100 g/L cells and a yield of 0.30 g/g.
- **required:** the inverse problem. It finds the titer needed for \$10, \$20 and \$50/kg at each production time.
- **physical limit:** a product yield of 0.50 g/g (82% of the energetic ceiling), the lowest measured maintenance, no growth during production, and 400 m³ vessels. $O_2$ transfer is capped at 250 mol/m³/h, which assumes head pressure, and heat removal at 40 kW/m³. Sugar is at contract price (\$0.25/kg) and the plant makes 100 kt/yr. Specific productivity is at most 40 mg/g/h. A grid search over cell density, production time and titer finds the cheapest feasible design.

| Grade | Meaning |
|---|---|
| A | Measured, primary source or official statistic (peer-reviewed data, EIA, Eurostat, BLS, USDA) |
| B | Design value or patent example from a named source (NREL, Intelligen, Humbird, patents) |
| C | Company claim, press release, industry survey, secondary transcription |
| D | Derived by analogy or calculation from graded inputs |
| E | Placeholder guess; flagged for replacement |

Table 3: Evidence grades (`data/evidence/README.md`).

**Uncertainty.** Two tools are used:

- **Tornado.** Each parameter with a graded range is moved to its low and high value, one at a time.
- **Monte Carlo.** Triangular draws are taken over each scenario's uncertain parameters. Draws that break a physical limit are kept in the record but excluded from the percentiles, which therefore describe feasible plants only. The feasible fraction is reported alongside.

## 3 Calibration

The structure was frozen first. Each anchor then changes only its own process inputs, prices, cost year and financing. Misses are recorded in `docs/VALIDATION.md` before anything changes. `tests/test_calibration.py` asserts each one inside a documented band, so a change that moves a result cannot pass silently.

| Anchor (type) | Quantity | Ours | Theirs | Diff. |
|---|---|---|---|---|
| 1. Humbird baker's yeast (software + reality) | cost, $/kg dry cells (2018) | 1.890 | 1.890 | +0.00% |
| 2. NREL 2011 on-site cellulase (consistency) | glucose, kg/kg protein | 4.26 | 4.17 | +2.2% |
| | $O_2$, mol/kg protein | 83.1 | 84 | −1.1% |
| | cost, $/kg protein (2007), Humbird capital | 11.61 | 4.24 | +174% |
| | cost, NREL's own capital convention | 6.05 | 4.24 | +43% |
| 3. Intelligen 2025 OVA (consistency) | cost, $/kg powder, Humbird convention | 47.82 | 43.52 | +10% |
| | total capital investment | $147M | $92M | +60% |
| | glucose, kg/kg powder | 9.4 | 13.0 | −28% |
| | electricity, kWh/kg powder | 11.3 | 27.1 | −58% |
| 4. Voutilainen 2021 recombinant case (consistency) | sugar, kg/kg protein | 6.37 | 6.15 | +3.6% |
| | cost, €/t protein, their DCF, Humbird capital | 25,381 | 9,007 | +182% |
| | cost, their fixed costs + light capital | 12,356 | 9,007 | +37% |

Table 4: Calibration anchors. Anchor 1 is Humbird's own workbook, ported line by line; anchors 2–4 are other models run on their own inputs. Details and diagnoses: `docs/VALIDATION.md`.

**What the anchors say.**

1. **The port is exact.** It reproduces Humbird's baker's-yeast cost to 0.004%. That cost sits on the market-implied $1.90/kg, so the method is anchored to a real price (with the caveat that the 33% margin is Humbird's assumption).
2. **The mass and energy balances agree with all three published PF and enzyme designs.** Sugar agrees within 4% and $O_2$ within about 1% of NREL. Electricity is lower than theirs. Our aeration is power-optimised; NREL sizes agitators for peak load, and Intelligen aerates at 1.2–1.6 vvm and runs a fertiliser section.
3. **Total costs differ by convention, not by biology.** On the same process, the published costs are 2.7–2.8× lower than ours (NREL, Voutilainen). The differences are:
   - vendor-quote fermenters instead of Aspen-based curves;
   - Lang factors of 2.7–4 instead of about 5;
   - fixed costs of 1.5–2% of capital per year instead of 9%.

   With their conventions our model still lands above theirs, by +37% (Voutilainen) and +43% (NREL). The remainder is mostly labour and buildings.
4. **Intelligen's total agrees** (+10%). Part of the agreement comes from differences that cancel: our capital is higher, and our glucose and electricity are lower. It is a weak pass.

Humbird's convention is the only one anchored to a real product price, so it is our base. It is also the more conservative. The light convention is shown everywhere as a named structural sensitivity, not as an error band. The NREL report itself notes that enzyme producers quoted costs about 47% above its estimate.

A port finding: Humbird's compressor formula gives about 1.6× the ideal adiabatic power at 2.1 bar. We keep it for fidelity to anchor 1. The effect on PF costs is about $0.1/kg.

# 4 Results

## 4.1 Base case

| Cost item, $/kg protein | BLG | OVA | Physical intensity, per kg BLG | |
|---|---|---|---|---|
| capital charge (14.6%/yr) | $12.28 | $13.45 | glucose | 12.7 kg |
| maintenance | $3.37 | $3.69 | oxygen | 7.7 kg |
| insurance and local tax | $4.22 | $4.62 | electricity | 14.5 kWh |
| glucose | $5.09 | $5.06 | natural gas (drying, steam) | 34 MJ |
| labour | $1.37 | $1.36 | spent cells (dry) | 3.2 kg |
| electricity | $1.25 | $1.38 | overall yield on glucose | 0.079 g/g |
| media, ammonia, DSP chemicals | $0.73 | $0.73 | fermenter volume | 601 m³ per kt/yr |
| utilities other, membranes | $0.86 | $1.30 | staff | 11.3 per kt/yr |
| **$MSP_{CCF}$** | **$29.18** | **$31.60** | total capital investment | $843M |
| per kg powder | $23.56 | $25.52 | production fermenters | 31 × 200 m³ |

Table 5: Base case: **K. phaffii**, 30 g/L, 72 h production, 10 kt/yr, US Midwest, Humbird capital convention.

Capital dominates the base case. The capital charge, maintenance and insurance together are 68% of the BLG price. Glucose is 17%.

- **Why so much capital.** A 10 kt/yr plant at 30 g/L needs 31 production fermenters of 200 m³. Every batch occupies a vessel for about 104 h, including turnaround, and yields 4,358 kg of protein.
- **Why glucose is modest.** The overall yield of protein on glucose is only 0.079 g/g: the cells, their maintenance and the product reaction all consume sugar. Even so, at $0.40/kg glucose costs less than a fifth of the price.

This is the first lever the rest of the paper explores: the product made per m³ of fermenter per hour.

### 4.2 Q1: cost against titer, with every public and claimed data point

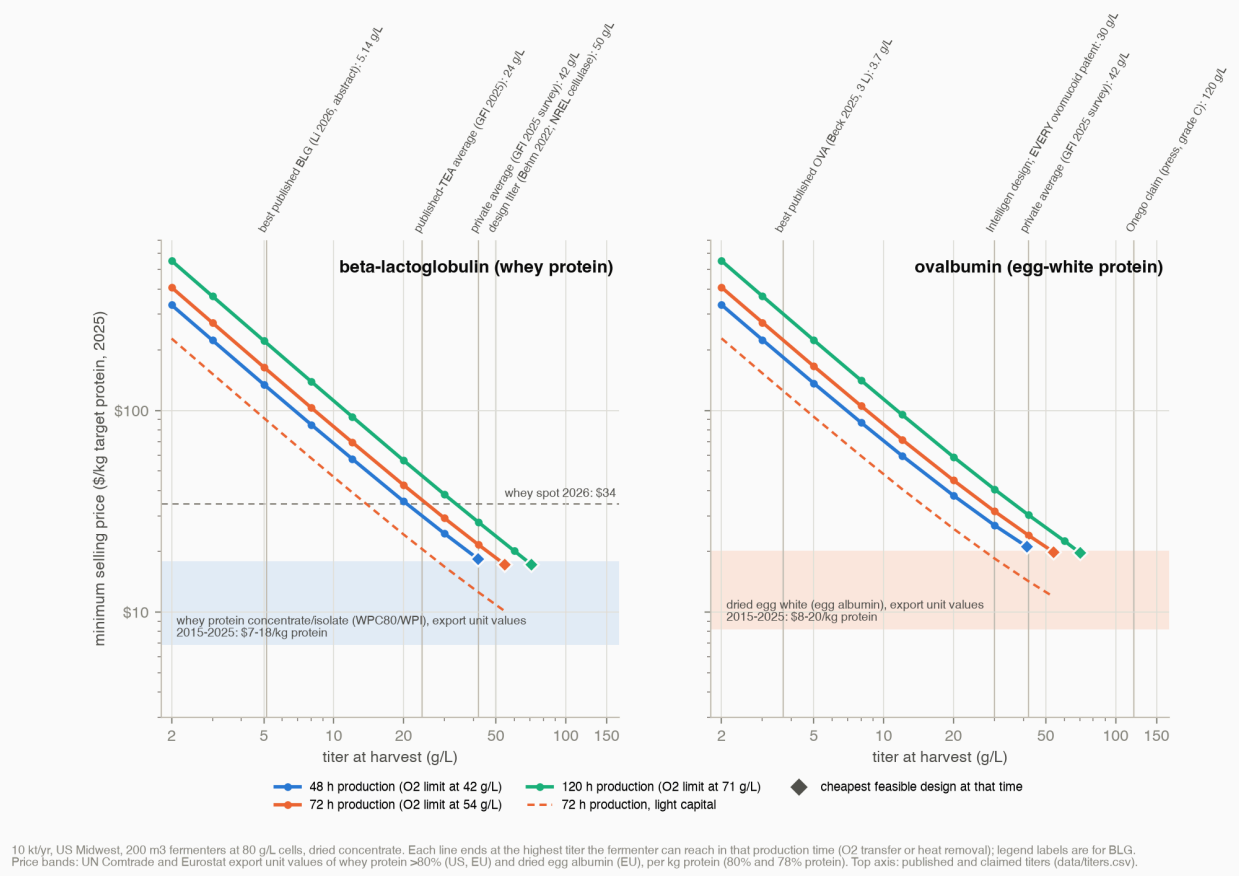



Figure 2: Minimum selling price against harvest titer, for three production times. Vertical lines mark the best public titers, the published-TEA and private-sector averages (GFI 2025), and design or claimed titers. Lines end where the $O_2$ or heat limit of the fermenter is exceeded.

Cost falls roughly as 1/titer until the $O_2$ wall (Figure 2):

- **Best public academic strains:** BLG at 5.14 g/L costs $188/kg; OVA at 2 g/L in **T. reesei** costs $467/kg. That is an order of magnitude above both incumbents.
- **Private-sector average (42 g/L):** $22.8 (BLG) and $25.2 (OVA), or $13.3 and $14.9 with light capital.

**The incumbents.** We use official trade unit values, not quotes, because they form a consistent 2015–2025 series:

- **Whey protein >80%** (WPC80 and WPI mixed; US and EU exports): $7–$18 per kg protein.
- **Dried egg albumin** (EU exports): $8–$20.
- **2026 whey spike.** WPC80 and WPI prices rose to about $34 per kg protein at the 2026 peak (USDA Dairy Market News), a record year driven by protein demand. At that price both the base case ($29.2) and the industry benchmark ($22.8) would undercut whey. At 2015–2025 prices neither would with Humbird capital; with light capital the benchmark ($13.3) is inside the band.

Parity at production cost is necessary but not sufficient for commercial parity. PF proteins also compete on functionality and on price per unit of function.

### 4.3 Q2: what drives the cost

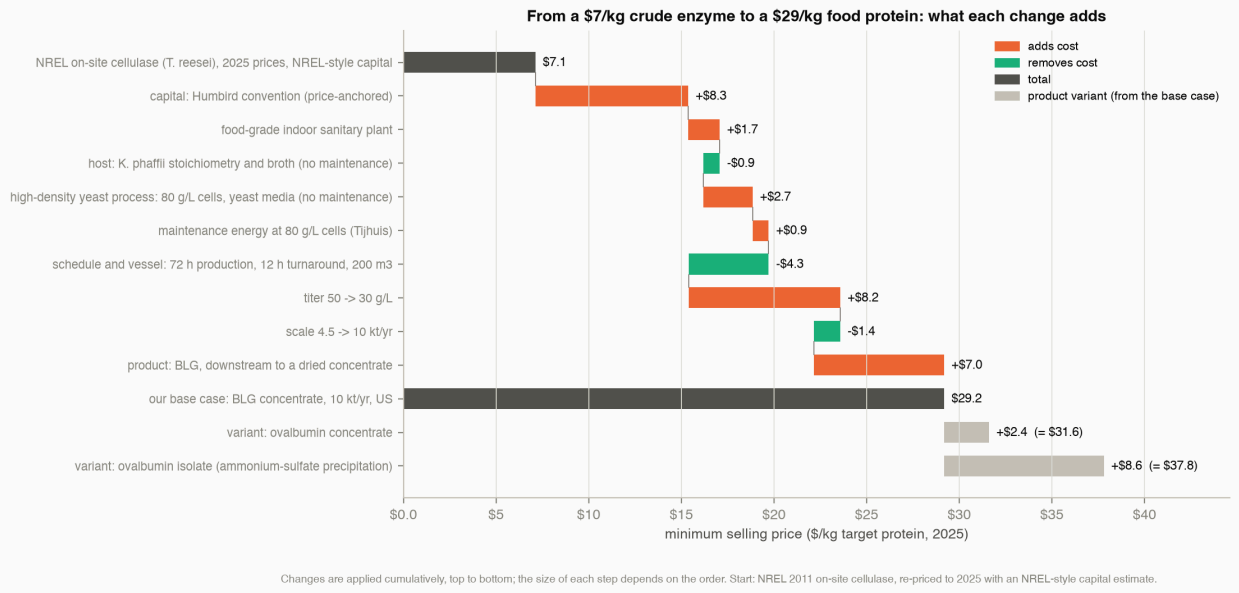



Figure 3: From NREL's on-site cellulase to our base case, one change at a time. The start is NREL's 2011 design re-priced to 2025 with an NREL-style capital estimate.

**Why a food protein costs more than an industrial enzyme.** Figure 3 walks from an NREL-style crude enzyme ($7.1/kg) to our base BLG concentrate ($29.2/kg). The largest steps are:

- **The capital convention:** +$8.3.
- **The titer drop from 50 to 30 g/L:** +$8.2.
- **Downstream processing and drying:** +$7.0. NREL's enzyme is used as whole broth.
- **Two savings:** moving to a fast yeast process with a 12 h turnaround (–$4.3), and scale (–$1.4).
- **The ovalbumin isolate variant** adds precipitation and a second recovery loss: +$8.6 over the BLG concentrate.

The order of the steps changes their sizes but not the total.

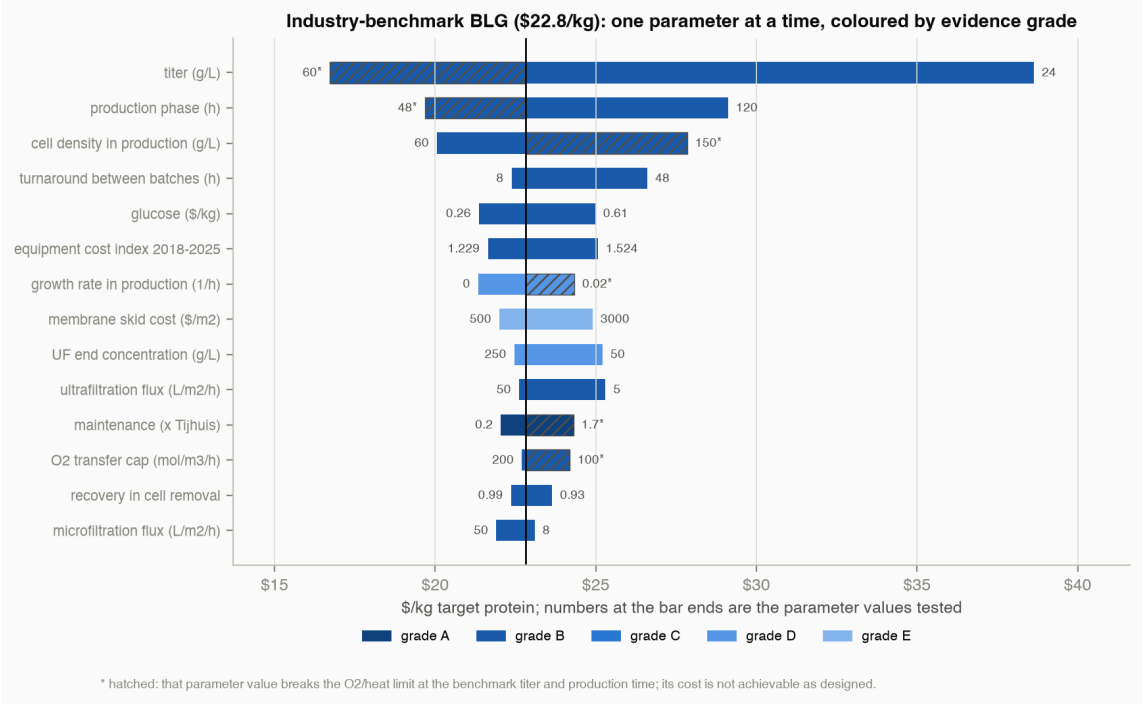



Figure 4: One-at-a-time sensitivity around the industry benchmark (BLG, 42 g/L), coloured by the evidence grade of each parameter's range. Hatched bars break a physical limit.

**Tornado** (Figure 4). Around the industry benchmark, the largest swings come from:

1. titer (g/L): $16.7–$38.6 (grade B)
2. production phase (h): $19.7–$29.1 (grade B)
3. cell density in production (g/L): $20.1–$27.8 (grade B)
4. turnaround between batches (h): $22.4–$26.6 (grade B)
5. glucose ($/kg): $21.4–$25.0 (grade B)
6. equipment cost index 2018-2025: $21.6–$25.1 (grade B)

Strain performance leads: titer, production time and cell density. They set how much protein a fermenter makes per hour, and so how much capital each kg carries. Turnaround time, the sugar price and the equipment cost index follow.

Two levers matter less than one might expect:

- **Product yield on sugar** (0.20–0.45 g/g) moves the price by only $0.7 (rank 18). Most of the sugar makes and maintains the cells, not the product, so cell density drives sugar use more than yield does. A low yield matters more for oxygen: at 0.20 g/g the benchmark breaks the $O_2$ limit.
- **Vessel size** (100–500 m³) moves it by $0.6 (rank 19). Humbird's fermenter cost curve and installation factors leave little economy of scale per m³ beyond 200 m³.

Few strain-performance ranges are grade A. They rest on company-reported averages and design assumptions, because no public BLG or OVA fed-batch data report titer, rate and yield together. The membrane skid cost is still a grade-E placeholder, yet ranks 8: it is the most valuable single number to source next.

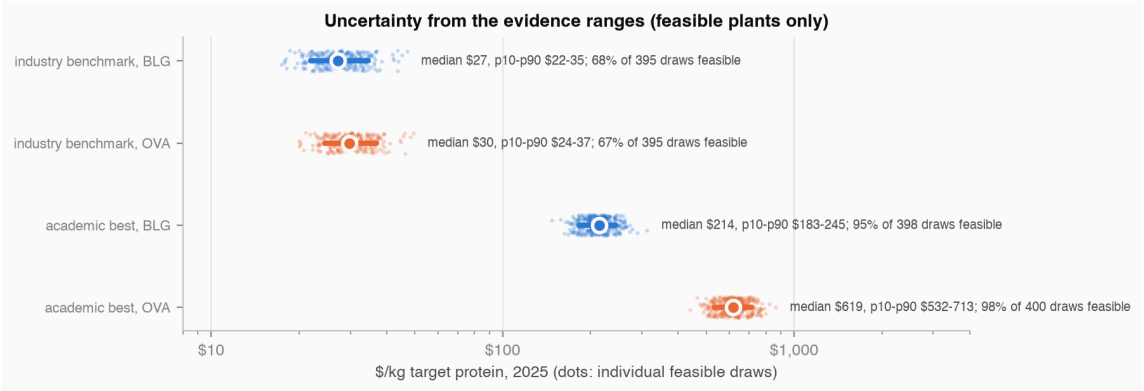



Figure 5: Monte Carlo over each scenario's evidence ranges (triangular), feasible plants only.

**Monte Carlo** (Figure 5). Over the evidence ranges, the industry benchmark's median is $27 for BLG (p10–p90 $22–$35). Only 68% of the draws are feasible: combinations of high titer and short production time break the $O_2$ limit. For the academic strains the whole distribution sits above $183/kg.

## 4.4 Q3: what must be true for $10 and $20 per kg

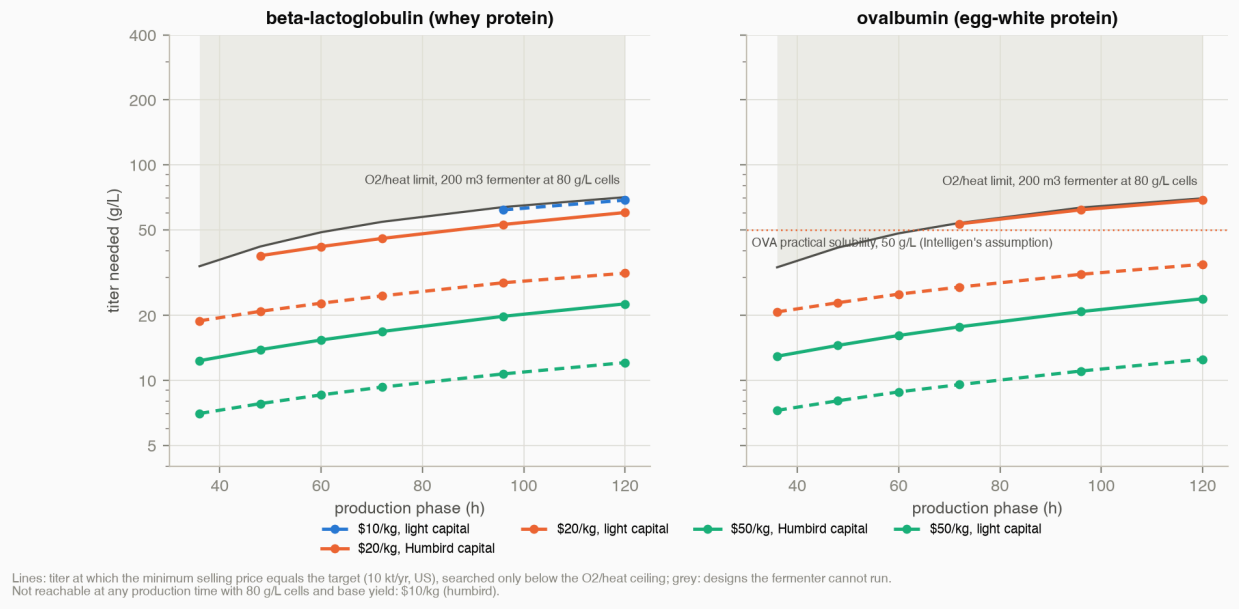



Figure 6: Titer needed for a target price, against production time. Grey: designs the fermenter cannot run ($O_2$ or heat limit, 80 g/L cells).

| Production phase (h) | 36 | 48 | 60 | 72 | 96 | 120 |
|---|---|---|---|---|---|---|
| $O_2$/heat ceiling, BLG | 34 | 42 | 49 | 54 | 64 | 71 |
| $10/kg, Humbird | — | — | — | — | — | — |
| $20/kg, Humbird | — | 38 | 42 | 46 | 53 | 60 |
| $50/kg, Humbird | 12 | 14 | 15 | 17 | 20 | 23 |
| $10/kg, light | — | — | — | — | 62 | 69 |
| $20/kg, light | 19 | 21 | 23 | 25 | 28 | 31 |
| $50/kg, light | 7 | 8 | 9 | 9 | 11 | 12 |

Table 6: BLG titer (g/L) needed for each target price, searched only below the $O_2$/heat ceiling (first row). — means the target is not reachable at that production time in a 200 m³ vessel at 80 g/L cells and base yield. OVA needs slightly more. With Humbird capital, $20/kg of OVA needs at least 53 g/L, above OVA's assumed practical solubility of 50 g/L (`results/analyses.json`, key `required`).

Figure 6 and Table 6 answer the inverse question:

- **$50/kg** needs 12–23 g/L, depending on production time. That is 2.4–4.4× the best public academic BLG titer, and well within today's industrial claims.
- **$20/kg** needs about 38–46 g/L in 48–72 h under the Humbird convention. That is at or above the private-sector average and close to the ceiling. The light convention needs only 21–25 g/L.
- **$10/kg is not reachable with Humbird capital** at the base yield and cell density: the titer it would need lies above the $O_2$/heat ceiling at every production time. With light capital it needs 62–69 g/L in 96–120 h, just under the ceiling.

Titer alone is therefore not enough for $10/kg. The other levers are yield, cell density, production time, vessel size, the capital convention and scale; the physical-limit scenario pulls them all at once.

## 4.5 Q4: the physical-limit floor

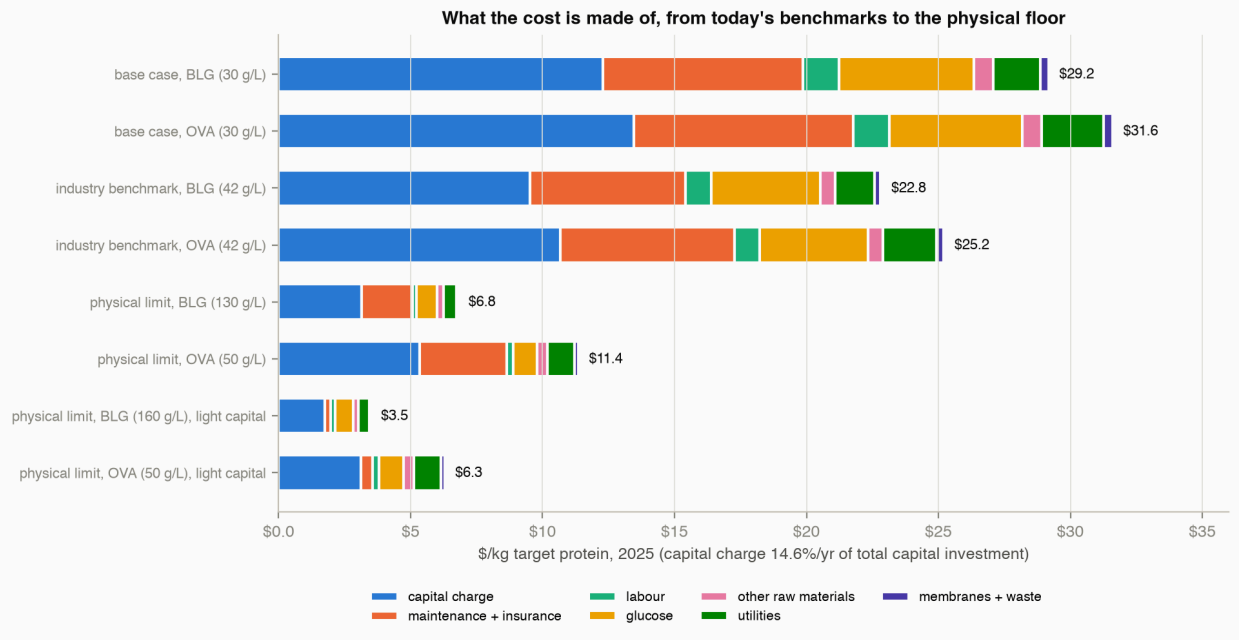



Figure 7: Cost structure from today's benchmarks to the physical limit. The best academic strain ($188/kg BLG) is off this scale.

The physical-limit search (Figure 7) picks the cheapest feasible design. The best BLG design has 130 g/L cells, a 72 h production phase and 130 g/L. Its specific productivity is 33 mg/g/h against the 40 mg/g/h cap.

The floor is $6.8/kg BLG with Humbird capital and $3.5 with light capital. For OVA it is $11.4 and $6.3.

- **Capital still dominates at the floor.** Capital-related costs are 75% of the BLG floor. Glucose ($0.78) is the largest non-capital item.
- **Why OVA's floor is higher.** OVA is held to its practical solubility of 50 g/L, in both the broth and the ultrafiltration retentate. That limit is Intelligen's assumption, not a measurement. If OVA could be concentrated like BLG, its floor would be $6.8 ($3.6 light). The OVA floor is therefore mostly a statement about that one number.

The floor is not a forecast. It is what physics and stoichiometry allow once every biological parameter is at its best measured or theoretical value, all at the same time. The BLG floor is at the bottom edge of the 2015–2025 whey band ($6.8–$17.8). The OVA floor is inside the egg-white band.

## 4.6 Q5: company claims against the physics

| Claim or benchmark | Titer | Modelled as | Shortest feasible production phase | Implied $q_p$ (mg/g/h) | MSP, Humbird | MSP, light |
|---|---|---|---|---|---|---|
| GFI 2025 private-sector average | 42 g/L | *K. phaffii*, 80 g/L cells | 49 h | 15 | $18.5 | $11.1 |
| GFI 2025 published-TEA average | 24 g/L | *K. phaffii*, 80 g/L cells | 24 h | 15 | $24.1 | $14.8 |
| Behm 2022 BLG design titer (T. reesei) | 50 g/L | *T. reesei*, 35 g/L cells | 39 h | 48 | $13.7 | $8.1 |
| T. reesei homologous secreted protein (Behm 2022, grade C) | 100 g/L | *T. reesei*, 35 g/L cells | 128 h | 49 | $13.1 | $7.3 |
| Onego ovalbumin claim (press, grade C) | 120 g/L | *T. reesei*, 35 g/L cells | 251 h | 48 | $16.8 | $8.7 |

Table 7: Claimed titers checked against $O_2$ transfer and heat removal in a 200 $m^3$ fermenter: the shortest production phase in which each titer is reachable, the specific productivity that then implies, and the cost there. The check is physical only. Whether a strain can sustain that productivity is a separate, biological question; our base case assumes about 7 mg/g/h.

Of the 5 claims and benchmarks, 5 can be reached within the $O_2$ and cooling limits if the production phase is long enough (Table 7). The price of a high titer is time, and time is fermenter capacity, so a claimed titer alone does not settle the cost.

Physics does not rule the claims out. Reaching them as fast as physics allows, though, implies specific productivities of 15–49 mg per g of cells per hour, against 7 in our base case. The **T. reesei** claims need the highest. At 35 g/L cells, a fungus must secrete about 49 mg/g/h for 10 days to reach 120 g/L. Whether a strain can sustain that is the open question, and no public data answer it.

### 4.7 Q6: scale and siting

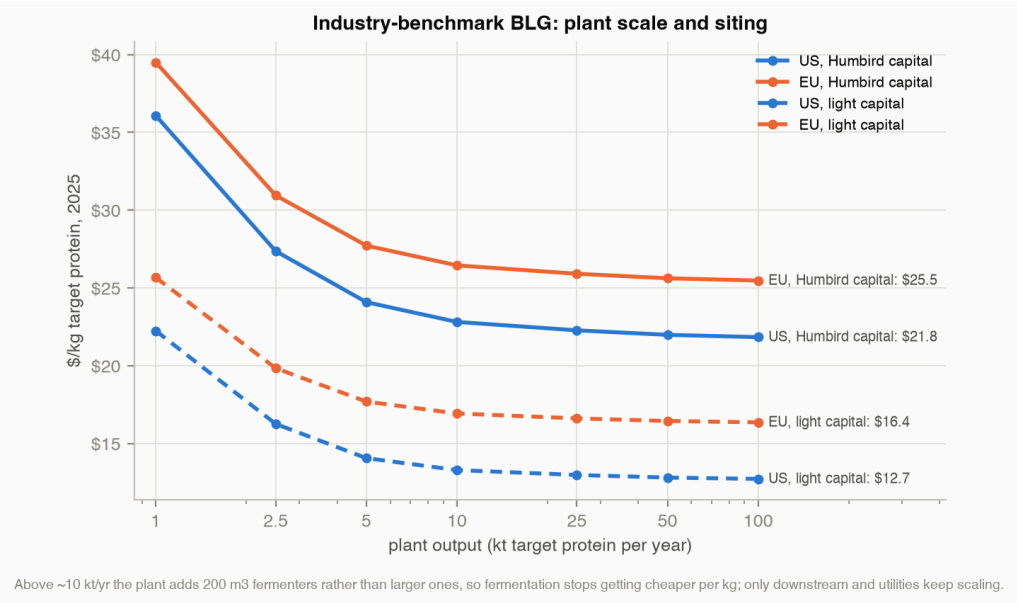



Figure 8: Industry-benchmark BLG against plant output, US Midwest and north-west Europe, both capital conventions.

**Scale** (Figure 8):

- Going from 1 to 10 kt/yr cuts the price by $13.3.
- Going from 10 to 100 kt/yr saves only a further $1.0.
- The reason: beyond about 10 kt/yr the plant adds more 200 m³ fermenters rather than bigger ones. Only the downstream train and the utilities keep scaling.
- Larger vessels help little: 100 to 500 m³ moves the benchmark by $0.6/kg (Figure 4).

**Siting.** A north-west Europe plant costs $4.4/kg more than the US Midwest in the base case. The difference comes from:

- glucose: $0.65 against $0.40 per kg;
- electricity: $0.146 against $0.086 per kWh;
- gas: $16.1 against $4.8 per GJ.

Labour is slightly cheaper in the Netherlands on our basis. Construction costs are assumed equal (Section 6).

## 5 Discussion

### 5.1 Answers, in brief

- **Q1.** Today's public strains make BLG and OVA at $188 and $467 per kg: not competitive. The private-sector average titer gives $22.8–$25.2, or $13.3–$14.9 with light capital. Against 2015–2025 prices:
  - with Humbird capital, BLG is above the whey band and OVA above the egg-white band;
  - with light capital, BLG is inside the whey band and OVA inside the egg-white band.
- **Q2.** Fermenter productivity (titer per hour of vessel time) and the capital convention drive the cost. Sugar and energy prices come second. The evidence for the strain parameters is the weakest link. It consists of

design values (grade B) and survey averages (grade C), because no public fed-batch dataset reports titer, rate and yield together.
- **Q3.** Under our base convention, $20/kg needs 38 g/L at 48 h or 46 g/L at 72 h, close to the $O_2$ ceiling. $10/kg needs more than titer.
- **Q4.** The physical floor is $6.8 (BLG) and $11.4 (OVA) per kg.
- **Q5.** 5 of 5 claimed titers are physically reachable in our fed-batch model, some only with long production phases, which cost capital.
- **Q6.** Scale matters up to about 10 kt/yr. EU siting adds about $4/kg.

### 5.2 Why published PF costs are lower

Published PF cost estimates for food proteins span roughly $12–39/kg:
- GFI Europe and Arthur D. Little (2025): $24–33/kg for their "P1" molecules today, and $12/kg for α-lactalbumin after improvements;
- Intelligen (2025): $39/kg for an OVA isolate at 1 kt/yr.

Where we can reproduce the process (NREL, Voutilainen), our calibration shows that the gap to our base is mostly accounting, not biology. Their fermenters are priced from vendor quotes, their installation and indirect factors are lighter, and their fixed costs are 1.5–2% of capital instead of Humbird's 9%.

Neither convention is "correct". Humbird's is anchored to the price of a real fermented commodity. The light one matches how the process industry quotes greenfield enzyme plants. A buyer of capital equipment in 2026 will probably land between them. We therefore read our base as a conservative central estimate and the light convention as an optimistic one, not as a confidence interval.

### 5.3 For funders

In order of expected cost reduction per unit of evidence gained:
1. **Public fed-batch data.** Titer, productivity and yield measured **together** at 1–10 m³ for BLG and OVA, in **K. phaffii** and **T. reesei**. No public dataset reports all three.
2. **Productivity, not only titer.** Faster production raises capacity per vessel. Near the $O_2$ wall, extra titer is only useful if it comes without extra time.
3. **Shared scale-up and capital-light plants.** Capital is the largest cost item in every scenario. Conversions of existing fermentation capacity (ethanol, enzymes) would move a plant toward the light convention.
4. **Downstream data.** Membrane skid costs (a grade-E placeholder, rank 8 in the tornado), ultrafiltration end concentrations and fluxes, and microfiltration flux (one patent example). After strain performance, these are the largest evidence gaps.

On the evidence here, PF BLG undercuts whey on production cost in spike years like 2026. At 2015–2025 prices it needs both the private-sector average titer and light capital ($13.3, inside the band). With Humbird's capital estimate, even the $O_2$ ceiling at 72 h gives $17.2, at the top edge of the band. PF OVA faces a similar gap. Egg-white prices spiked in 2022–2023 with avian influenza (EU export unit value up to $20/kg protein), which is when OVA's case is strongest.

> **Check (Pablo):** Author's judgment, not a model output (please revise): the probability that a PF plant built with 2026 technology makes BLG below $10/kg protein is about 0.15, with low confidence. The physical-limit case shows $10 is physically possible, but only if every lever moves at once.

### 5.4 For EU policy

At EU sugar, electricity and gas prices, the same plant costs $4.4/kg more than in the US Midwest (base BLG):
- US-level glucose would remove $3.2 of the gap.
- US-level energy prices would remove $1.3.
- Cheaper, longer capital does more than either. A capital charge at 4% over 20 years instead of 7.5% over 10 cuts $6.1, more than the whole transatlantic gap. At 4% over 10 years it cuts $1.9.

The infrastructure that matters most is not fermenters per se, but the combination of:
- cheap fermentable sugar next to the plant;
- chilled water and power for aeration;

- membrane and drying capacity.

The cost and form of capital matter more than siting. Instruments that lower the cost of capital, or enable conversions of existing plants, move the cost more than energy subsidies would.

### 5.5 What would change these conclusions

- A peer-reviewed BLG or OVA fed-batch at ≥40 g/L in ≤48 h with a yield of ≥0.3 g/g would put $20/kg inside the Humbird-convention feasible region.
- Evidence that production-scale $O_2$ transfer routinely exceeds 200 mol/m³/h (pressurised vessels) would move the wall up.
- Real capital costs of recent PF plants (for example Liberation Labs, 600 m³) would settle the convention question.

## 6 Limitations

- **Strain parameters.** Only a few are grade A. Yields and specific productivities are design assumptions (Intelligen) or survey averages (GFI). The academic titers come from abstracts; 15 titer rows are not yet verified in full text (`data/titers.csv`).
- **OVA solubility** of 50 g/L is Intelligen's practical assumption. If OVA stays soluble higher, the OVA ceiling rises.
- **Heat removal** above 114 m³ needs external exchangers; the 25 kW/m³ cap is grade D. No source gives a production-scale ceiling.
- **Construction costs** are the same in the EU and the US. No sourced location factor is on disk.
- **Continuous processes** and cell recycle are not modelled. Nor are methanol-induced expression and chromatography for high-purity isolates.
- **Spent cells** are costed at zero (no credit, no disposal cost; grade E, range −$50 to +$100 per t).
- **Humbird's compressor formula** overstates compressor power by 1.6×. We keep it for fidelity; the effect is about $0.1/kg.
- **Grade-E inputs** (placeholders) remain for: `hosts.treesei.mu_max_per_h`, `media.antifoam_kg_per_m3`, `dsp.membrane_skid_usd_m2`, `utilities.biomass_disposal_usd_t_dry`, `capital.dsp_area_factor`, `capital.lab_m2`. See Appendix A.

## 7 Scoring the pre-registered priors

The priors were written in `docs/SCOPE.md` before any model output existed.

| Prior (2026-09-26) | Stated | Outcome | Score |
|---|---|---|---|
| 1a. Intelligen configuration, $/kg OVA (80% interval) | $38–60 | $50.8/kg OVA | inside |
| 1b. Base case, $/kg protein (80% interval) | $28–50 | $29.2 (BLG), $31.6 (OVA) | inside, near the low end |
| 1 (superseded). Purified BLG/OVA at 30 g/L | $20–40 | as above | inside |
| 2. P(physical floor < $10/kg) | 0.65 | BLG $6.8 (yes); OVA $11.4 (no) | Brier 0.12 / 0.42 |
| 3. P(a public BLG/OVA yield on sugar turns up) | 0.20 | none found (an LCA inventory value only) | Brier 0.04 |
| 4. P(first-try calibration within tolerance): yeast / NREL / Intelligen | 0.6 / 0.6 / 0.5 | yes / no (+136%) / yes (−3.6%) | Brier 0.16 / 0.36 / 0.25 |
| 5. P(public v1 by 2027-04-30 / 2027-07-31) | 0.5 / 0.8 | open | — |

Table 8: Pre-registered priors and outcomes. Prior 4 is scored on the first run (2026-09-26), before the evidence updates recorded in `docs/VALIDATION.md`.

The interval priors were well placed but wide. The base case landed near the low end of both. The calibration prior was about right on average; the NREL miss came from capital conventions, which the prior did not anticipate.

# 8 Reproducibility

```
.venv/bin/python -m pytest                      # 22 tests: Humbird port, calibration bands
.venv/bin/python scripts/run_analyses.py        # all results -> results/analyses.json
.venv/bin/python scripts/make_figures.py        # docs/figures/*.png
typst compile --root . report/pf_tea_report.typ # this document
```

Inputs are in `data/pf/base.yaml` (every value with a range, grade and source), with scenario overrides in `data/pf/scenarios/`. The evidence files are `data/evidence/*.yaml`, with verbatim quotes and locators. Calibration cases are in `data/calibration/`.

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# A Parameters (generated from `data/pf/base.yaml`)

| Parameter | Value | Range | Gr. | Source |
|---|---|---|---|---|
| plant.product | blg | | | blg \| ova |
| plant.spec | concentrate | | | concentrate (GRN 1056-type, UF/DF + dry) \| isolate (adds precipitation) |
| plant.annual_target_protein_t | 10000 | 1000–100000 | C | gfi2025costs Table 2: private-sector volumes 2,500-25,000+ t/yr; SCOPE base |
| plant.operating_days | 330 | 320–350 | B | NREL 2011 uses 8,410 h/yr (96%); Humbird A.1 350 d; Intelligen ~342 batches x 24 h; 330 d allows for campaigns and cleaning |
| plant.location | us_midwest | | | us_midwest \| eu_nw |
| strain.host | kphaffii | | | |
| strain.titer_g_L | 30 | 5–60 | B | intelligen_ovalbumin2025 design basis 30 g/L; Every2024-HY ovomucoid 30.19 g/L (patent Ex. 9); gfi2025costs public avg 24, private avg 42 |
| strain.t_prod_h | 72 | 48–120 | B | intelligen_ovalbumin2025 p.6: expression phase 72 h; NREL 2011: 96 h production |
| strain.X_prod_g_L | 80 | 40–150 | B | intelligen_ovalbumin2025 p.7: 92 g/L DCW at harvest; K. phaffii glucose fed-batch 93-169 g/L (Shemesh 2024, A); patents 100-200 (Every2024-HY, B). data/evidence/hosts.yaml |
| strain.X_max_g_L | 150 | 100–200 | A | K. phaffii glucose fed-batch 93-169 g/L (Shemesh 2024); typical 150; patents up to 200 (hosts.yaml). Growth stops here |
| strain.mu_prod_per_h | 0.01 | 0–0.02 | D | K. phaffii production-phase growth rates (evidence: data/evidence/hosts.yaml); Intelligen expression phase makes 8 g cells per 28 g product |
| strain.Y_ps_g_g | 0.28 | 0.2–0.45 | B | intelligen_ovalbumin2025 Eq.(2): 100 glucose -> 28 rOVA (+8 cells, 4 HCP); NREL 2011 cellulase: 90% of carbon via protein reaction, 0.24 g/g overall |
| strain.X0_g_L | 5 | 2–15 | D | inoculum ~5% v/v from seed at ~100 g/L (Intelligen: 5% inoculum, 3 seed stages); Humbird A.1 starts at 15 g/L |
| hosts.kphaffii.biomass_formula | (table in file) | | A | Carnicer et al. 2009 Microb Cell Fact 8:65, Table 4: Fab-producing K. phaffii CH1.665N0.134O0.602S0.0039 (evidence hosts.yaml; re-check table cell) |
| hosts.kphaffii.mu_max_per_h | 0.2 | 0.15–0.25 | A | hosts.yaml: mu_max ~0.2 1/h on sugar vs 0.11-0.12 on methanol |
| hosts.kphaffii.temperature_C | 30 | | B | intelligen_ovalbumin2025 p.6: 30 C |
| hosts.kphaffii.maint_factor | 1 | 0.2–1.7 | A | multiplier on Tijhuis 1993 maintenance (= 0.0176 g/g/h, Humbird A.1 method); measured K. phaffii m_s 0.0023-0.029 g/g/h (Nieto-Taype 2020, Rebnegger 2016, Maurer; hosts.yaml) |
| hosts.kphaffii.kla_factor | 1 | | B | yeast broth treated as non-coalescing water-like (Humbird Fig1.3) |
| hosts.kphaffii.cell_removal | centrifuge | | | |
| hosts.treesei.biomass_formula | (table in file) | | B | humbird2011 p.38: T. reesei cell mass CH1.645O0.445N0.205S0.005 |
| hosts.treesei.mu_max_per_h | 0.1 | 0.05–0.2 | E | no sourced T. reesei mu_max found (hosts.yaml gap entry); fungi grow slower than yeast |
| hosts.treesei.temperature_C | 28 | | B | humbird2011 Table 19: 28 C |
| hosts.treesei.maint_factor | 1 | 0.5–2 | D | as for yeast |
| hosts.treesei.kla_factor | 0.5 | 0.3–0.8 | D | viscous mycelial broth lowers kLa vs water-like correlations (humbird2011 used a fungal-specific correlation); evidence file |
| hosts.treesei.cell_removal | filter_press | | | |
| fermenter.volume_m3 | 200 | 100–500 | B | engineering.yaml: industrial aerobic vessels 200-1,000 m3 (Humbird, B), lysine 500 m3 (B), baker's yeast 200 m3 (B); PF designs 136 m3 (Intelligen, B), 150 m3 (Liberation Labs, C), 80-120 m3 culture volume (GFI Europe, C); Humbird STRFERM quotes to 196 m3 (larger sizes extrapolate the cost curve) |
| fermenter.h_to_d | 2 | | B | Humbird Equip Table10 STRFERM 200,000: 196.5 m3, D 5 m |
| fermenter.fill_max | 0.8 | | B | Humbird Fig1.3!C8; NREL 2011 80% |

| Parameter | Value | Range | Gr. | Source |
|---|---|---|---|---|
| fermenter.pv_max_W_m3 | 5000 | 2000–5000 | B | engineering.yaml: Humbird 2021 p.5 'agitation rate up to ~5,000 W/m3' design limit in high-density fermentation (B); NREL 300 m3 T. reesei max 600 kW (~2,000 W/m3) |
| fermenter.us_max_m_s | 0.1 | 0.05–0.15 | D | flooding/foaming limit (evidence file) |
| fermenter.otr_cap_mol_m3_h | 150 | 100–200 | B | engineering.yaml: Humbird A.1!C93 design cap 150 (B); '200 mol O2/m3-h is about as high as it gets' (Humbird 2021 p.10, B); >120 needed by new processes (Takors, A); 100 industrial experience (C) |
| fermenter.heat_removal_max_kW_m3 | 25 | 15–40 | D | engineering.yaml: no source gives a production-scale ceiling. At OTR 150 the load is 19.2 kW/m3 metabolic (460 kJ/mol O2, A) + ~1.2 kW/m3 agitation (Humbird 200 m3 model, B); P. pastoris on methanol released 19.2 W/L at lab scale (A); jackets are insufficient above ~114 m3, so external chilled-water exchangers are assumed (C; U ~1.4 kW/m2/K, B). 25 = 1.2 x the OTR-150 load |
| fermenter.turnaround_h | 12 | 8–48 | B | intelligen_ovalbumin2025 (5 fermenters, 24 h cycle, 108 h culture -> ~12 h); Humbird A.1 8 h; NREL 48 h |
| fermenter.feed_glucose_wt | 0.5 | | B | intelligen_ovalbumin2025 p.6: 500 g/L glucose solution |
| fermenter.seed_ratio | 0.1 | | B | NREL 2011 p.39: seed vessels 10% of the next; Intelligen 5% inoculum |
| fermenter.seed_time_h | 24 | 15–36 | B | intelligen 15 h per seed step; NREL 2-day seed cycle incl. cleaning |
| fermenter.seed_final_X_g_L | 60 | | D | seed grown to high density before transfer |
| media.salts_kg_per_m3 | 38 | 20–50 | B | intelligen_ovalbumin2025 Table 1 basal medium: CaSO4 0.93 + MgSO4.7H2O 14.9 + KOH 4.13 + K2SO4 18.2 g/L (+H3PO4) = 38 g/L |
| media.salts_price_usd_kg | 0.2 | 0.15–0.4 | B | intelligen_ovalbumin2025 Table 3 'Basal Salts' $0.200/kg |
| media.h3po4_kg_per_kg_cells | 0.047 | | B | intelligen Eq.(1): 1.87 H3PO4 per 40 cells |
| media.h3po4_price_usd_kg | 0.85 | | B | intelligen Table 3: H3PO4 (85%) $0.850/kg |
| media.antifoam_kg_per_m3 | 0.5 | 0.1–1 | E | no dose found; NREL corn oil 0.1% v/v (~0.9 kg/m3) as the high end; guess |
| media.antifoam_price_usd_kg | 1.12 | 0.5–2 | C | prices.yaml: Voutilainen 2021 design value EUR 1,000/t x 1.12 (no market price found); range is a guess |
| media.methanol_kg_per_kg_product | 0 | | | methanol-free base; Intelligen uses 0.28 kg/kg (Table 3: 280 t per 1,000 t). Cost only: its O2 and heat are not added to the stoichiometry |
| media.methanol_price_usd_kg | 0.91 | | B | intelligen_ovalbumin2025 Table 3: $0.910/kg |
| dsp.recovery_cell_removal | 0.97 | 0.93–0.99 | B | intelligen_ovalbumin2025: two disk-stack centrifuges + MF with retentate recycle; overall train 0.857 incl. 0.884 purification (A1_SOLUTION 2.1) -> clarification ~0.97 |
| dsp.dilution_to_g_L_cells | 45 | | B | intelligen p.7: broth diluted to 45 g/L biomass before centrifugation |
| dsp.centrifuge_stages | 2 | | B | intelligen p.7 |
| dsp.centrifuge_hours_per_batch | 22 | | B | intelligen p.8: centrifugation 22 h (semi-continuous) |
| dsp.mf_flux_LMH | 10 | 8–50 | B | dsp.yaml: EVERY US2025/0134130 MF polishing 'Average flux (expected) 10 +/- 2 LMH' (B); microalgae MF 20-100 (Slegers, D) |
| dsp.mf_recovery | 0.98 | | D | retentate recycled (intelligen p.7) |
| dsp.uf_flux_LMH | 30 | 5–50 | B | intelligen p.8: 30 L/m2/h; Petrides 20-50; EVERY patent DF at 50 g/L '5 +/- 2 LMH' (dsp.yaml) |
| dsp.uf_rejection | 0.995 | | B | intelligen p.8: rejection 0.995 for rOvalbumin |
| dsp.uf_end_g_L | 150 | 50–250 | D | BLG: whey UF routinely concentrates to high solids; OVA capped at 50 g/L practical solubility (intelligen p.8) |
| dsp.diafiltration_volumes | 3 | 2–6 | D | salt and sugar washout to meet the protein spec (evidence file) |
| dsp.membrane_price_usd_m2 | 150 | 50–200 | B | dsp.yaml: Petrides 2015, industrial-biotech membranes 'under $200/m2' |
| dsp.membrane_life_y | 0.5 | 0.25–2 | B | dsp.yaml: Petrides 2015, industrial membranes last >2000 operating hours (lower bound) |

| Parameter | Value | Range | Gr. | Source |
|---|---|---|---|---|
| dsp.membrane_skid_usd_m2 | 1200 | 500–3000 | E | PENDING evidence (dsp.yaml) |
| dsp.membrane_kWh_m3_permeate | 5 | 1–13 | D | dsp.yaml: UF ~13 kWh/m3 permeate (A, upper end); typical crossflow UF lower |
| dsp.precip_recovery | 0.891 | | B | intelligen p.8: 90% precipitated x 99% retained by basket centrifuges |
| dsp.ammonium_sulfate_kg_per_m3 | 350 | | B | intelligen p.8: 350 g/L |
| dsp.ammonium_sulfate_usd_kg | 0.18 | 0.18–0.33 | B | intelligen Table 3: $0.180/kg; EU import UV 2025 EUR 0.19 (A); US import UV $0.33 (A, high) |
| dsp.ammonium_sulfate_recycled_frac | 0 | | | Intelligen sells it on as fertiliser; base assumes no recovery credit |
| dsp.dryer_inlet_solids | 0 | | | dryer is fed the UF retentate as is (no evaporator); dairy WPC80 feeds ~36% solids after evaporation (Tetra Pak, dsp.yaml), EVERY ~5% |
| dsp.dryer_MJ_per_kg_water | 4.87 | 3.4–6 | A | dsp.yaml: Baker & McKenzie survey average 4.87 GJ/t water evaporated; milk powder 3.4 (3-stage) to 4.9 (1-stage), Ramirez et al. |
| dsp.dryer_recovery | 0.98 | | D | cyclone + bag filter losses (evidence file) |
| dsp.concentrate_target_frac_dry | 0.85 | 0.7–0.92 | D | target protein share of dry powder after UF/DF of a secreted protein; GRN 1056 minimum is 0.56 (0.70 x 0.80) |
| dsp.isolate_target_frac_dry | 0.97 | | B | intelligen p.9: 99% purity (dry basis) |
| dsp.powder_moisture | 0.05 | | B | intelligen p.9: 5% moisture; GRN 1056 max 7% |
| dsp.wet_cells_solids | 0.25 | | D | centrifuge sludge ~25% dry solids |
| utilities.electricity_usd_kWh | US 0.0862 EU 0.1456 | 0.072–0.1014 | A | prices.yaml: EIA industrial average 2025 US 8.62 c/kWh (Iowa 7.20 = low, Illinois 10.14 = high); EU = Eurostat nrg_pc_205 NL band IF (70-150 GWh/yr, the size of a 10 kt plant) 2025 mean EUR 0.1289/kWh excl. VAT x ECB 1.13 |
| utilities.natural_gas_usd_GJ | US 4.78 EU 16.13 | 3.34–6.38 | A | prices.yaml: EIA US industrial 2025 $5.23/Mcf / 1.036 MMBtu/Mcf (low = Henry Hub $3.52/MMBtu, high = Illinois $7.05/Mcf); EU = Eurostat nrg_pc_203 EU27 band I4 (0.1-1 PJ/yr) 2025 mean EUR 14.27/GJ GCV x 1.13 |
| utilities.glucose_usd_kg | US 0.4 EU 0.65 | 0.26–0.61 | B | prices.yaml: GFI/Hawkwood 2025 Midwest DE95 $0.40/kg (B); low = Humbird contract $0.26 (2019, B); high = IMARC US dextrose Jun 2025 $611/t (C). EU (grade C, proxies): median of Comext glucose-syrup export UV 2025 EUR 0.43/kg as traded at ~75% DS ($0.65/kg DS), EU white sugar 2025 EUR 537/t ($0.61), IMARC Germany Jun 2025 $674/t |
| utilities.ammonia_usd_kg | 0.61 | 0.3–0.86 | B | prices.yaml: US anhydrous NH3 import unit value 2025 $0.611/kg (UN Comtrade, A proxy; EU import UV EUR 0.53 = $0.60); low = Humbird SI $0.30 (2019); high = USDA Iowa retail 2025 $784/short ton |
| utilities.water_usd_m3 | US 1.5 EU 1.62 | 0.26–3 | B | prices.yaml: GFI/Hawkwood 2025 Midwest $1.50/m3 (B); low = Humbird well water $0.26 (2019); high = Intelligen $3/t; EU = VEMW 2025 NL large-user average EUR 1.43/m3 (C) |
| utilities.wastewater_usd_m3 | 1 | 0.4–4 | B | dsp.yaml: $0.2-0.5/m3 below 1 g/L BOD (Petrides); Chicago $0.42/kg BOD implies ~$4/m3 at 10 g/L (A) |
| utilities.steam_usd_t | US 16.9 EU 56.9 | 12–25 | D | gas-fired LP steam at Humbird's ratio: $9.7/t at $2.9/MMBtu gas = 3.53 GJ gas per t steam (Humbird SI RawUtil, B), times the gas price above; Intelligen $20/t (2025); range guess |
| utilities.biomass_disposal_usd_t_dry | 0 | −50–100 | E | spent yeast as feed/fertiliser credit or disposal cost; PENDING evidence |
| utilities.rt_refrig | 0.2629248030187212 | | B | Humbird RawUtil!BY4 (1.24 hp/ton, Couper) |
| utilities.cooling_water_dT_C | 8 | | B | Humbird A.1!C158 |
| utilities.building_W_m2 | 50 | | B | Humbird Equip Table9: Sanitary 50 W/m2 |
| labour.operator_salary_usd | US 60430 EU 55800 | 58320–79970 | A | prices.yaml: BLS OEWS May 2025 Iowa 51-9011 chemical equipment operators $60,430 (US mean $60,710; Humbird's basis was $50k in 2018); low = Iowa 51-8091 $58,320, high = US 51-8091 chemical plant operators $79,970. EU (grade B, derived): Eurostat SES 2022 NL ISCO OC8 gross earnings EUR |

| Parameter | Value | Range | Gr. | Source |
|---|---|---|---|---|
| | | | | 43,822 x NL manufacturing labour-cost growth 2022-24 (47.3/42.0) x ECB 1.13 |
| labour.burden | 1 | | B | Humbird A.1!C246: 100% labour burden |
| labour.shifts_per_position | 4.5 | | B | Humbird A.1!C225 |
| labour.employees_per_supervisor | 10 | | B | Humbird A.1!E242 |
| labour.supervisor_factor | 1.4 | | B | Humbird RawUtil!BG24 |
| labour.attention_production | 0.1 | | B | Humbird A.1!C231: 10% of online+offline time |
| labour.attention_seed | 0.1 | | B | Humbird A.1!C230 (large seed) |
| labour.worker_efficiency | 0.8 | | B | Humbird A.1!C224 |
| labour.continuous_fte_per_shift | (table in file) | | D | Humbird A.1 continuous-area FTE rules (0.25-1 per area) extended to the DSP areas |
| capital.cost_index_2018_to_2025 | 1.33 | 1.229–1.524 | B | prices.yaml: CEPCI 2018->2023 x1.323 (B) and 2025 = 1.016 x 2024 (publisher) -> ~x1.33-1.35; BLS PPI machinery & equipment WPU11 x1.327 (A). low = BLS PPI chemical manufacturing PCU325325 x1.229 (A), the index Humbird's Equip sheet uses (290.2 in 2018); high = PPI pumps & compressors WPU1141 x1.524 (A). Equipment-specific PPIs run higher still: heavy-gauge tanks x1.92, heat exchangers x1.64 |
| capital.basis | humbird | | | humbird (base; ACCE fermenter quotes + Petrides/ Humbird factors, anchored to baker's-yeast price) \| nrel (vendor-quote fermenters, NREL 2011 installation + Table 27 indirects, 3% ISBL maintenance + 0.7% FCI insurance): structural sensitivity |
| capital.nrel_basis | (table in file) | | B | humbird2011 App. A: F-400 $400,500 + A-400 $580,000 (2009$, x PPI 290.2/224.7), exponent 0.7 (Area 300 fermentors), install 2.0/1.5; Table 27: warehouse 4% + site 9% + piping 4.5% of ISBL; prorateable 10% + field 10% + home office 20% + contingency 10% + other 10% of TDC; Table 31 fixed costs |
| capital.facility_class | commodity_aerobic | | | commodity_aerobic (food-grade, indoor sanitary) \| commodity_anaerobic (outdoor, lysine-type) |
| capital.facility_classes | (table in file) | | B | Humbird Equip Table14 (Petrides classes, as tabulated by Humbird): ISBL non-bioreactor DC x PEC, OSBL DC x PEC, E&C x TDC, fees & contingency x TPC, fermentation-area building class |
| capital.missing_frac | 0.1 | | B | Humbird 4FB sheets: 10% missing-capital estimate on ISBL and OSBL |
| capital.packing_factor | 0.15 | | B | Humbird 4FB-020!D232: fraction of area occupied by fermenters |
| capital.building_usd_m2_level_2018 | (table in file) | | B | Humbird Equip Table9 (ACCE 2016$ escalated to 2018) |
| capital.dsp_area_factor | 0.5 | | E | DSP + drying building area as fraction of fermentation area (PENDING review) |
| capital.office_m2_per_employee | 15 | | B | Humbird 4FB-020!D235 |
| capital.lab_m2 | 400 | | E | cell bank + QC labs for a food plant (guess) |
| finance.ccf_rate | 0.075 | | B | humbird2021 p.39: i = 7.5%, n = 10 y -> CCF ~15% TCI/ y |
| finance.ccf_life_y | 10 | | B | humbird2021 p.39 |
| finance.maintenance_frac_tci | 0.04 | | B | humbird2021 p.46 |
| finance.insurance_tax_frac_tci | 0.05 | | B | humbird2021 p.46 |
| finance.dcf_irr | 0.1 | | B | humbird2011 Table 33 |
| finance.dcf_life_y | 30 | | B | humbird2011 Table 33 |
| finance.tax_rate | 0.21 | | A | US federal corporate rate since 2018 (NREL 2011 used 35%) |
| finance.depreciation_years | 7 | | B | humbird2011 Table 33: 7-y MACRS (200% DB) for general plant |
| finance.construction_years | 3 | | B | humbird2011 Table 33: 8/60/32% |
| finance.working_capital_frac_fci | 0.05 | | B | humbird2011 Table 33 |
| finance.equity_frac | 1 | | | unlevered DCF for the MSP (NREL 2011 used 40% equity) |

# B Evidence base

| Evidence file | A | B | C | D | E | Total |
|---|---|---|---|---|---|---|
| data/evidence/hosts.yaml | 62 | 19 | 3 | 4 | 8 | 96 |
| data/evidence/dsp.yaml | 7 | 72 | 5 | 9 | 0 | 93 |
| data/evidence/engineering.yaml | 38 | 37 | 7 | 1 | 0 | 83 |
| data/evidence/prices.yaml | 62 | 23 | 12 | 0 | 0 | 97 |

Table 9: Evidence entries by grade. Each entry carries a source, a locator and, where a document was read, a verbatim quote.